Two View Geometry

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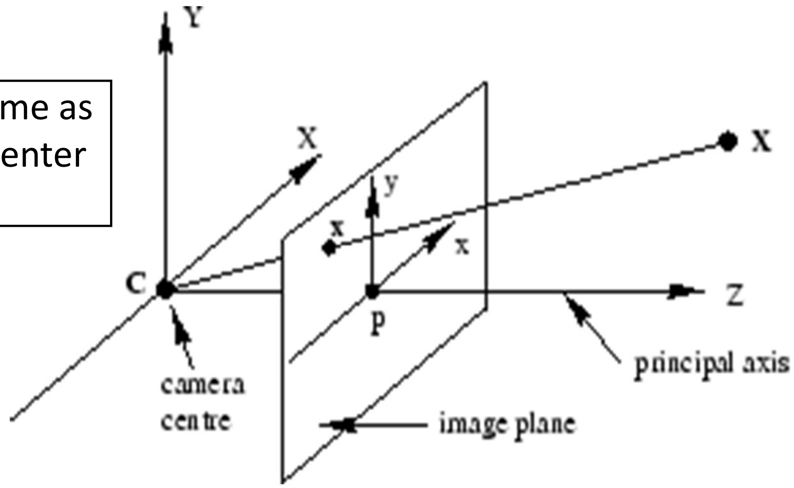
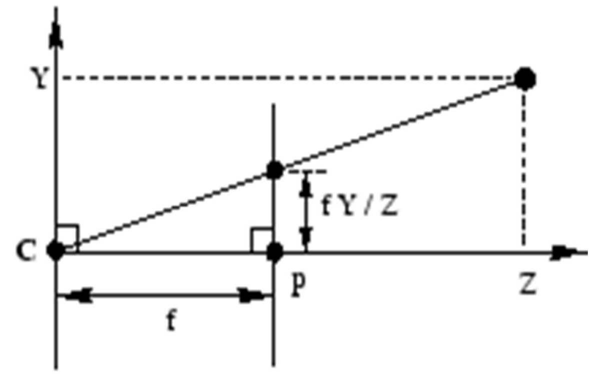
Pinhole Camera Model

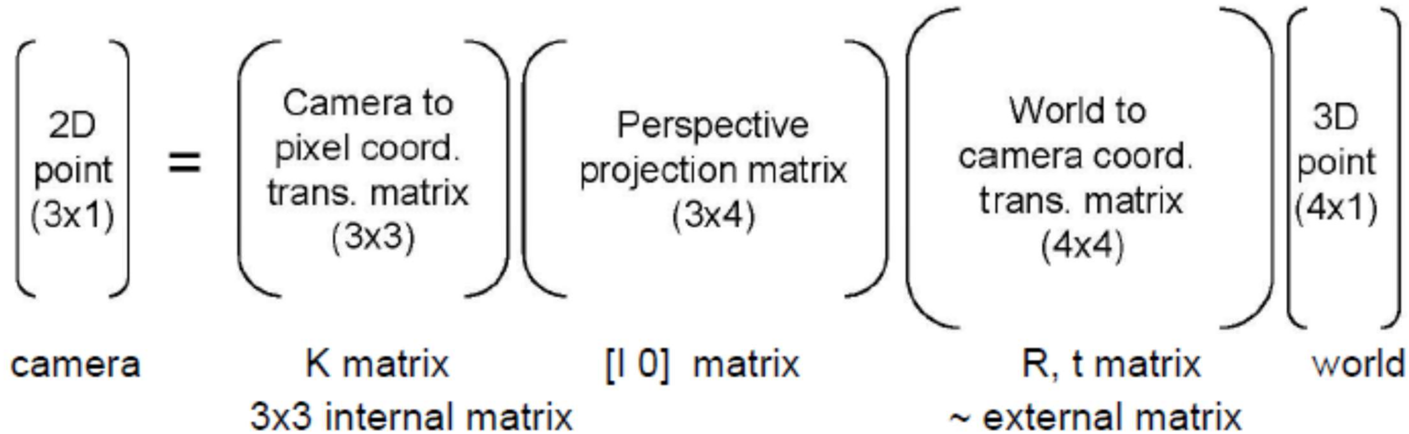
Camera Center is the same as the focal point, or the center of projection (‘**C**’)





- **Principal Axis** – Line from the camera centre perpendicular to the image plane.
- **Principal point** – point where the principal axis intersects the image plane ($z=f$)
- **Pixel coordinate frame** – Origin (0,0) is in the corner of an image; units are in pixels.

Image Formation Summary



- Rotation and Translation (3D world to 3D camera)
- Perspective Projection (camera 3D to image plane 2D)
- Scaling and Shifting (image plane 2D to pixel 2D)
- Use of homogeneous coordinates

Camera Parameters

- Intrinsic
  - Focal length (2-dof) f_x and f_y
  - Principal point (2-dof) c_x, c_y
  - Skew factor (1-dof) s
- Extrinsic
  - Rotation (3-dof) - $\mathbf{R}$
  - Translation (3-dof) – $\mathbf{t}$
- Total degrees of freedom: 3 + 3 + 2 + 2 + 1 = 11
  - Need 11 equations to estimate these parameters
  - Direct Linear Transform
  - Calibration toolboxes

Homogeneous Coordinates

- Multiply the coordinates by a non-zero scalar and add an extra coordinate equal to that scalar. For example,

$$(x, y) \to (x \cdot z, y \cdot z, z) \quad z \neq 0$$

$$(x, y, z) \to (x \cdot w, y \cdot w, z \cdot w, w) \quad w \neq 0$$

- Back to Cartesian coordinates

$$(x, y, z) \quad z \neq 0 \to (x / z, y / z)$$

$$(x, y, z, w) \quad w \neq 0 \to (x / w, y / w, z / w)$$

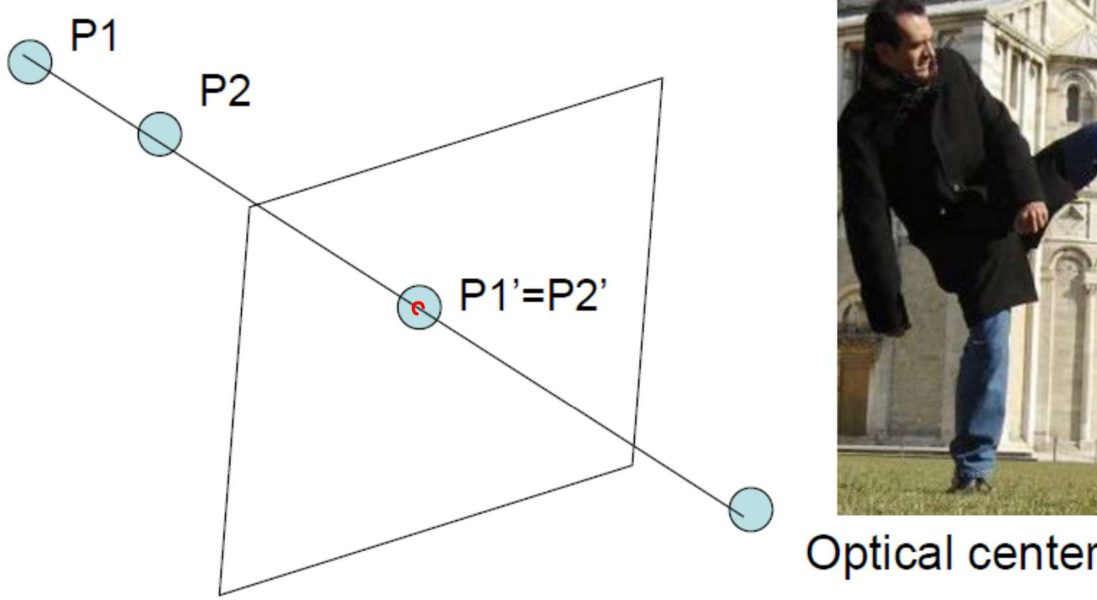
Why Homogeneous Coordinates?

- Unified representation for all points (including points at infinity – $[x,y,0]^T$)

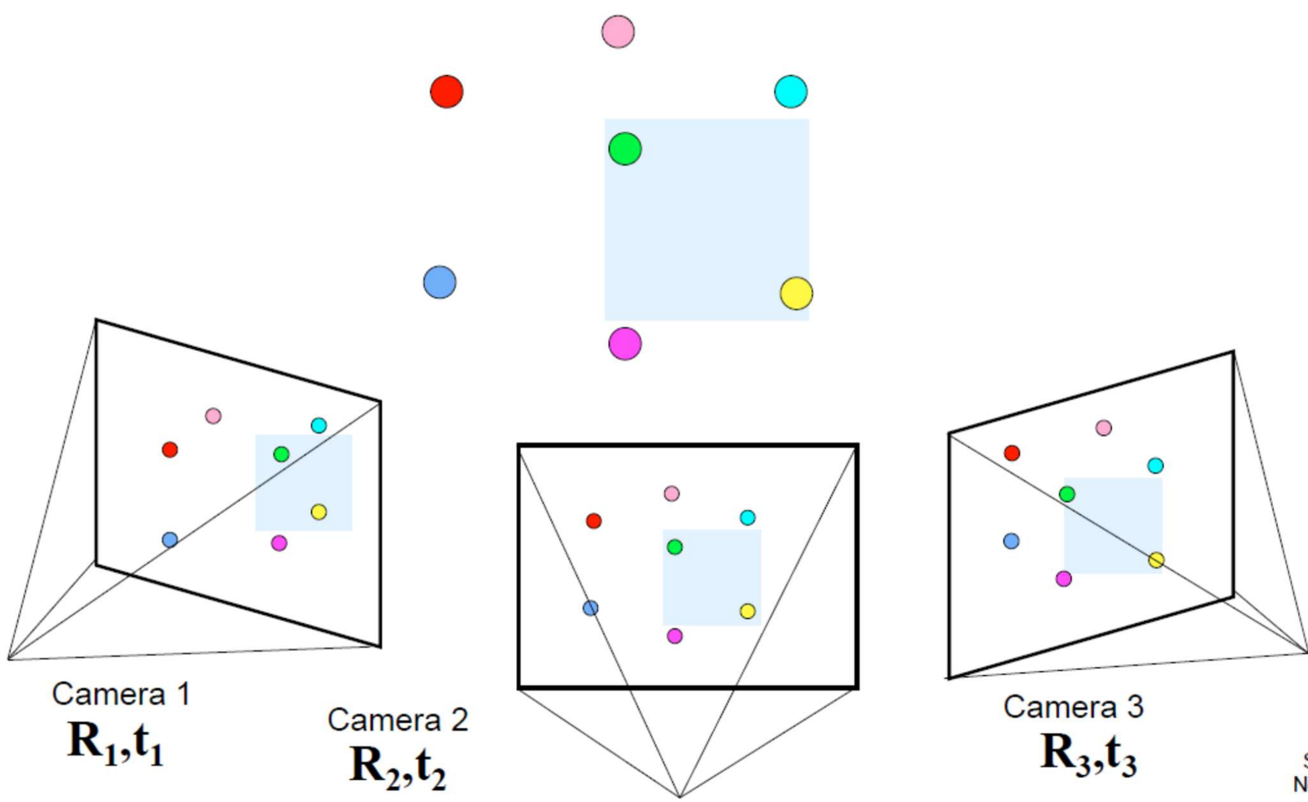
- Projective transformations are represented by *linear transformations* of homogeneous coordinates
  - Convenient!

Why multiple views?

- Structure and depth are inherently ambiguous from single views.

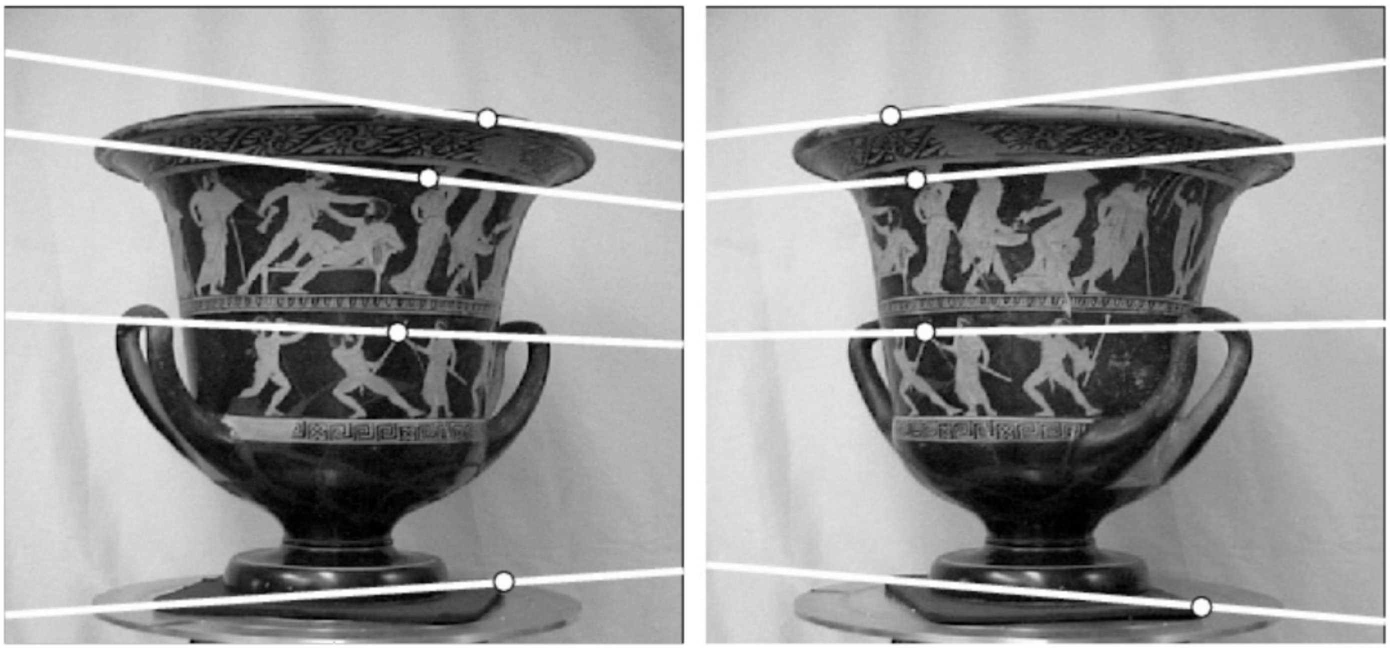


Multi-view geometry problems



Slide credit: Noah Snavely

Two-view geometry

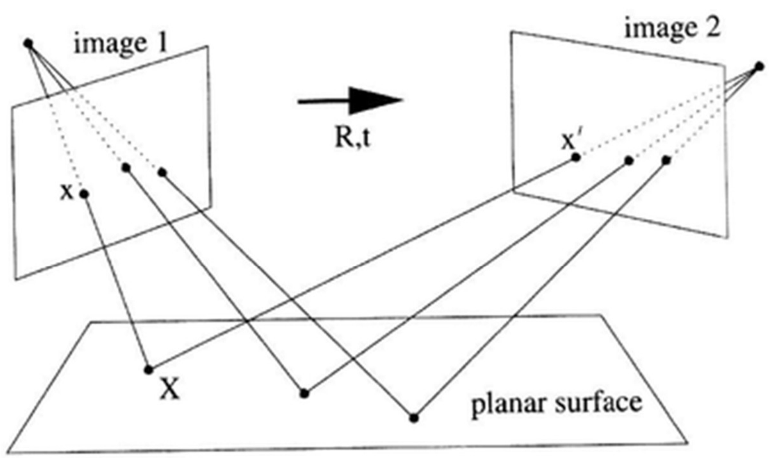


Planar Homography in Two Views

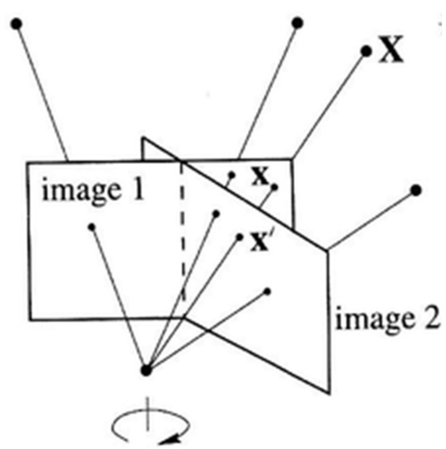
- Planar Scenes
- Pure Rotation

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^\top & b \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \mathbf{H}_p \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

2D Projective Transformation
(Planar Homography)



a



b

Geometric Transformations

- Isometries or Euclidean Transformation
- Similarity Transformation
- Affine Transformation
- Projective Transformations

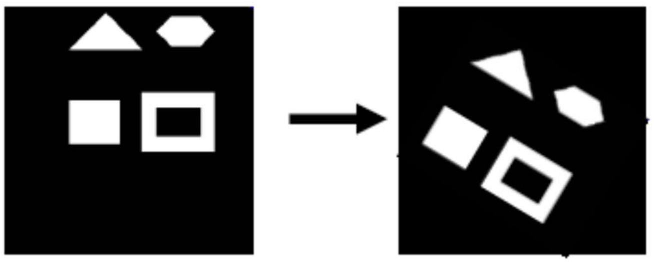
Transformation in 2D

Isometries:

[Euclideans]

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = H_e \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

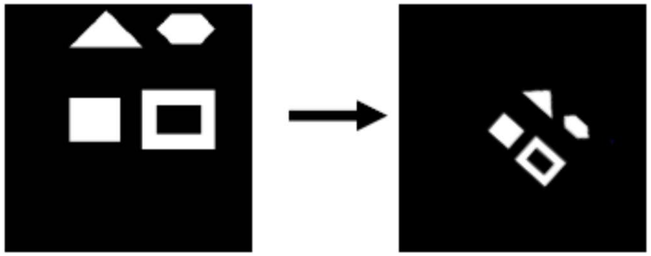
- Preserve distance (areas)
- ? DOF
- Regulate motion of rigid object



Transformation in 2D

Similarities:
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s\,R & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = H_s \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Preserve
    - ratio of lengths
    - angles
- ? DOF

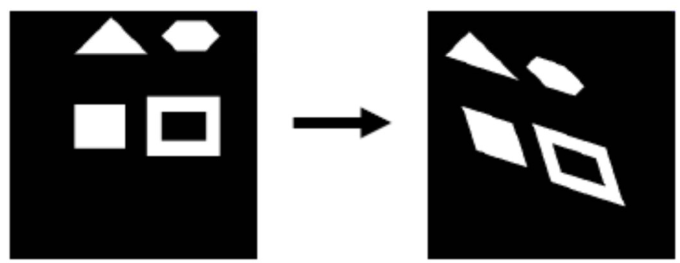


Transformation in 2D

Affinities:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = H_a \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \overbrace{R(\theta)}^{U} \cdot R(-\phi) \cdot \overbrace{D}^{\Sigma} \cdot \overbrace{R(\phi)}^{V^T} \qquad D = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$



Transformation in 2D

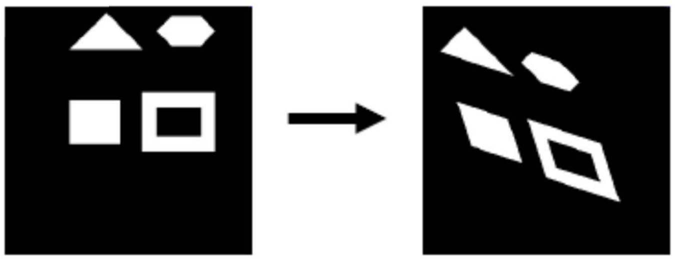
Affinities:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = H_a \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = R(\theta) \cdot R(-\phi) \cdot D \cdot R(\phi) \quad D = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

-Preserve:
- Parallel lines
- Ratio of areas
- Ratio of lengths on collinear lines
- others…

- ? DOF



Transformation in 2D

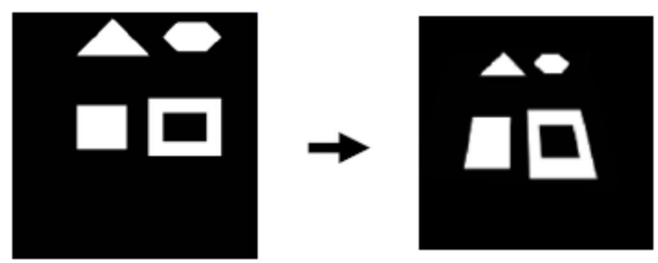
Projective:

$$\begin{bmatrix} z\,x' \\ z\cdot y' \\ z\cdot w' \end{bmatrix} = \cancel{z}\begin{bmatrix} A & t \\ \boxed{v} & b \end{bmatrix}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = H_p \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} \cancel{z}x'/zw' \\ z\,y'/zw' \end{bmatrix}$$

- ? DOF
- Preserve:
  - cross ratio of 4 collinear points
  - collinearity
  - and a few others…



2D Projective Transformation

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^\top & b \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \mathbf{H}_p \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$w' = v_1 \cdot x + v_2 \cdot y + b \cdot 1$$

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} \rightarrow \begin{bmatrix} x'/w' \\ y'/w' \end{bmatrix}$$

2D Affine/Similarity/Euclidean Transformation

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \mathbf{H}_{a/s/e} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$w' = 0 \cdot x + 0 \cdot y + 1 \cdot 1 = 1$$

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} \rightarrow \begin{bmatrix} x'/w' \\ y'/w' \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

Transformation in 2D

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^\top & b \end{bmatrix} \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} = \mathbf{H}_p \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$$

$$w' = v_1 \cdot x + v_2 \cdot y + b \cdot 0 \neq 0$$

$$\begin{bmatrix} x \\ y \\ 0 \end{bmatrix} \xrightarrow{\mathbf{H}_p} \begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} \rightarrow \begin{bmatrix} x'/w' \\ y'/w' \end{bmatrix}$$